

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2183

Woodenshoe 138 kV
September 22, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2183. The Interconnection Customer proposes a 207.9 MW battery energy storage facility interconnecting at the Woodenshoe 138 kV substation in the Maryland area. The request was received on September 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 459 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,221,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

TWINBROOK – TILLATOBA 138kV: 155.6% loading (252 MVA) following the outage of the WOODENSHOE – KENSINGTON 138kV; DFAX = 61.2%.

VEIRS MILL – TILLATOBA 138kV: 146.4% loading (245 MVA) following the outage of the KENSINGTON – WOODENSHOE 138kV; DFAX = 61.2%.

For the loss of WOODENSHOE – KENSINGTON 138kV, the VEIRS MILL – WOODENSHOE 138kV line loads to 139.8% of its 237 MVA rating. Project DFAX: 61.2%.

The KENSINGTON – TWINBROOK 138kV Ckt 1 is overloaded to 132.5% of its 287 MVA emergency rating for the loss of the TWINBROOK – KENSINGTON 138kV. The project contributes 5.1% DFAX (10.7 MW) to this overload.

The TWINBROOK – KENSINGTON 138kV Line 2 is overloaded to 132.5% of its 287 MVA emergency rating for the loss of the KENSINGTON – TWINBROOK 138kV L. The project contributes 5.1% DFAX (10.7 MW) to this overload.

For the loss of TWINBROOK – KENSINGTON 138kV Circuit 2, the TWINBROOK – KENSINGTON 138kV #3 loads to 132.5% of its 287 MVA rating. Project DFAX: 5.1%.

For the loss of AIR FORCE BASE – MONTGOMERY VILLAGE 138kV Ckt 1, the MONTGOMERY VILLAGE – AIR FORCE BASE 138kV #2 loads to 125.1% of its 438 MVA rating. Project DFAX: 5.2%.

AIR FORCE BASE – MONTGOMERY VILLAGE 138kV Cir. 1: 123.3% loading (445 MVA) following the outage of the MONTGOMERY VILLAGE – AIR FORCE BASE 138kV Cir. 2; DFAX = 5.2%.

The POLK – BELLS MILL 138kV is overloaded to 115.6% of its 266 MVA emergency rating for the loss of the PARKLAWN – TWINBROOK 138kV. The project contributes 6.0% DFAX (12.6 MW) to this overload.

For the loss of TILLATOBA – TWINBROOK 138kV L, the WOODENSHOE – KENSINGTON 138kV line loads to 112.5% of its 323 MVA rating. Project DFAX: 71.9%.

TIGER BAY – HUNTING HILL 138kV: 104.0% loading (287 MVA) following the outage of the AIR FORCE BASE 345/138kV XFMR; DFAX = 5.2%.

For the loss of TWINBROOK – KENSINGTON 138kV #2, the KENSINGTON – NIH 138kV loads to 103.0% of its 220 MVA rating. Project DFAX: 5.8%.

LINDEN SUBSTATION – KENSINGTON 138kV Ckt 1: 102.7% loading (534 MVA) following the outage of the KENSINGTON – LINDEN 138kV line; DFAX = 12.4%.

KENSINGTON – LINDEN 138kV: 102.7% loading (534 MVA) following the outage of the LINDEN – KENSINGTON 138kV Circuit 1; DFAX = 12.4%.

TIGER BAY 345/138 kV Auto: 100.9% loading (549 MVA) following the outage of the AIR FORCE BASE 345 kV/138 kV TX; DFAX = 5.2%.

BALD MTN – SUMMIT LAKE 138kV L: 100.9% loading (194 MVA) following the outage of the COLESVILLE – DARIEN 138kV; DFAX = 5.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TWNBROOK – TILLATOBA 138kV	Rebuild line with higher-capacity conductor	\$1,215,000	100% (\$1,215,000)
VEIRS MILL – TILLATOBA 138kV	Rebuild line with higher-capacity conductor	\$1,154,000	100% (\$1,154,000)
VEIRS MILL – WOODENSHE 138kV line	Rebuild line with higher-capacity conductor	\$1,613,000	100% (\$1,613,000)
TWINBROOK – KENSINGTON 138kV #1	Rebuild line with higher-capacity conductor	\$5,291,000	100% (\$5,291,000)
TWINBROOK – KENSINGTON 138kV Line 2	Rebuild line with higher-capacity conductor	\$5,371,000	100% (\$5,371,000)
TWINBOOK – KENSINGTON 138kV #3	Rebuild line with higher-capacity conductor	\$4,272,000	100% (\$4,272,000)
MONTGOMERY VILLAGE – AIR FORCE BASE 138kV #2	Rebuild line with higher-capacity conductor	\$870,000	100% (\$870,000)
AIR FRCE BASE – MONTGOMERY VILLAGE 138kV	Rebuild line with higher-capacity conductor	\$1,182,000	100% (\$1,182,000)
POLK – BELLS MILL 138kV	Rebuild line with higher-capacity conductor	\$1,193,000	100% (\$1,193,000)
KENSINGTON – WOODENSHOE 138kV	Rebuild line with higher-capacity conductor	\$7,369,000	100% (\$7,369,000)
TIGER BAY – HUNTING HILL 138kV line	Reconductor line	\$820,000	100% (\$820,000)
KENSINTON – NIH 138kV	Reconductor line	\$834,000	100% (\$834,000)
LINDEN SUBSTATION – KENSINGTON 138kV Ckt 1	Reconductor line	\$819,000	54% (\$439,000)
KENSINGTON – LINDEN 138kV	Reconductor line	\$485,000	54% (\$260,000)
TIGER BAY 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$14,592,000	100% (\$14,592,000)
BALD MTN – SUMMIT LAKE 138kV L	Reconductor line	\$746,000	100% (\$746,000)
	Total allocated to project		\$47,221,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.